

4(a)(i)	ester ;	1
4(a)(ii)	condensation ;	1
4(a)(iii)	1 (hydrocarbon chains of) different lengths / Y is longest / Z is shortest ; A correct numbers of carbon atoms in each fatty acid (X 18, Y 20 and Z 16) <i>if no ref. to different lengths</i> 2 Z saturated, X and Y unsaturated ; A only Z is saturated 3 X polyunsaturated, Y monounsaturated ; 4 X 3, Y 1, Z 0 double (C=C) bonds ; only Z only has single / C-C) bonds or (for whole molecule) X 4 double bonds , Y 2 double bonds, Z 1 double bond ;	3
4(b)(i)	any two from: 1 (hydrocarbon / fatty acid tails, are) non-polar / have no polar groups / not charged ; A equal sharing of electrons in molecule 2 not soluble / insoluble, (in water) ; 3 cannot form hydrogen bonds with water ; I repel water	2
4(b)(ii)	any two from: 1 no, hydrophilic / polar / phosphate, head / part, to, interact with / AW, water ; I triglycerides are hydrophobic 2 cannot form hydrogen bonds with water ; I repel water 3 cannot form a bilayer ; in water triglycerides, form micelles / form spheres / ball shaped / form globules / do not form thin films / AW ;	2